

Detailed Notes on Internal Truthfulness Detection in LLMs

Based on the paper:

The Internal State of an LLM Knows When It's Lying

(ArXiv: 2303.08774)

1 Background: Hallucination in Large Language Models

Large Language Models (LLMs) have demonstrated remarkable performance across a wide range of tasks, including question answering, reasoning, and text generation. Despite their success, LLMs are prone to generating factually incorrect statements while maintaining a confident and persuasive tone. This phenomenon is commonly referred to as *hallucination*. The danger of hallucination lies not only in incorrect information, but in the fact that users may trust these outputs due to the fluency and confidence with which they are presented.

Traditional evaluation methods often focus on surface-level correctness or rely on external fact-checking systems. However, these approaches do not answer a deeper question: whether the model itself internally distinguishes between true and false statements.

This motivates the central question of this work:

Do LLMs internally represent the truth or falsehood of a statement, even when they generate incorrect outputs?

2 Autoregressive Language Models: Correct Conceptual Understanding

2.1 Formal Definition of Autoregression

In machine learning, an autoregressive (AR) model is defined by its factorization of a joint probability distribution over a sequence:

$$P(x_1, x_2, \dots, x_T) = \prod_{t=1}^T P(x_t \mid x_1, x_2, \dots, x_{t-1})$$

This definition emphasizes **conditional dependence on the past**, not generation.

A common misconception is that autoregressive models must always use *previously generated tokens*. This is incorrect. Autoregression refers to *how probabilities are conditioned*, not where the tokens come from.

2.2 Teacher Forcing vs Autoregression

Teacher forcing refers to the training or inference strategy where the model is provided with ground-truth tokens as input at each time step.

Crucially:

- Autoregression describes the probability structure.
- Teacher forcing describes the source of input tokens.

Therefore, a model can be:

- Autoregressive **and** teacher-forced at the same time.

This is the standard training procedure for Transformer-based language models.

3 Training, Reading, and Generation: Three Distinct Modes

3.1 Training Mode

During training:

- The full sentence is known.
- The model predicts the next token at each position.
- Ground-truth tokens are provided as input (teacher forcing).
- A loss is computed and weights are updated.

Training uses autoregressive modeling with teacher-forced inputs.

3.2 Reading Mode (Inference-Time Teacher Forcing)

In the context of SAPLMA, *reading* refers to inference-time teacher forcing:

- A complete sentence is fed token-by-token.
- The model computes hidden states and next-token probabilities.
- No tokens are sampled or emitted.
- No loss is computed.

Architecturally, reading uses the **same Transformer decoder** with the same causal mask as training and generation.

3.3 Generation Mode

During generation:

- Only a prompt is given.
- The model predicts the next-token distribution.
- A token is sampled or selected.
- The generated token is fed back as input.

Generation introduces sampling noise and early commitment errors.

3.4 Summary Table

Mode	Autoregressive	Teacher Forcing	Sampling
Training	Yes	Yes	No
Reading	Yes	Yes	No
Generation	Yes	No	Yes

4 Why “Reading” Is Crucial for Studying Internal Belief

The claim that LLMs possess an internal belief about truthfulness must be demonstrated independently of generation artifacts.

If truthfulness can be detected:

- While the model is passive
- Without sampling
- Without committing to outputs

then the signal must be intrinsic to the internal representations.

Reading mode mirrors the training computation but eliminates confounding factors such as decoding strategy and error accumulation. This makes it the cleanest setting for probing internal belief.

5 Dataset Design: True–False Statements

5.1 Construction of the Dataset

The SAPLMA dataset consists of short, unambiguous factual statements labeled as **True** or **False**. Statements span six topics:

- Cities
- Animals
- Companies
- Chemical Elements
- Inventions
- Scientific Facts

False statements are created by swapping factual attributes between entities while maintaining grammatical correctness.

5.2 Why Topic-Wise Hold-Out Is Necessary

For each topic:

- The classifier is trained on the other five topics.
- The held-out topic is used only for testing.

This prevents the classifier from learning topic-specific heuristics and forces it to rely on internal truth signals encoded by the LLM.

This evaluation strategy is known as *leave-one-topic-out* probing.

6 SAPLMA Methodology

6.1 Hidden-State Probing

Each sentence is fed to a base LLM (e.g., OPT-6.7B or LLaMA2-7B). From a selected layer, the hidden-state activation vector is extracted (typically the final token).

These vectors serve as inputs to a lightweight classifier.

6.2 Classifier Training

For each held-out topic:

- One classifier architecture is trained.
- Training is repeated three times with different random initializations.
- Accuracy is averaged across runs.

In total, six topics and three runs yield eighteen trained classifiers, which are used only to compute evaluation statistics.

7 Sentence Probability vs Truthfulness

7.1 Language Model Probability

The probability assigned by an LLM to a sentence is given by:

$$P_{\text{LM}}(x_1, \dots, x_T) = \prod_{t=1}^T P(x_t \mid x_{<t})$$

This quantity reflects:

- Token frequency
- Syntax
- Sentence length

It does **not** directly measure factual correctness.

7.2 Limitations of Sentence Probability

Sentence probability is biased by:

- Common vs rare words
- Length normalization issues
- Fluency rather than semantics

As a result, fluent falsehoods may receive higher probabilities than awkward truths.

7.3 SAPLMA Probability

SAPLMA outputs a sigmoid score:

$$P_{\text{SAPLMA}}(\text{true} \mid h)$$

This score reflects whether the LLM’s internal representation aligns with truthfulness, independent of surface form.

8 Qualitative Evidence from Table 5

The qualitative examples demonstrate that SAPLMA:

- Is robust to sentence length
- Is insensitive to token frequency artifacts
- Handles typos gracefully
- Correctly identifies implicit truths

In contrast, sentence probabilities fluctuate widely due to syntactic and lexical factors.

These results strongly support the claim that truthfulness is encoded differently from linguistic likelihood.

9 Conclusion

This work addresses a fundamental limitation of Large Language Models (LLMs): their tendency to generate factually incorrect statements with high confidence. Rather than attempting to prevent hallucination through external knowledge injection or prompting strategies, the paper investigates whether LLMs themselves internally encode a notion of truthfulness.

The central contribution of this work is the introduction of **SAPLMA** (Statement Accuracy Prediction based on Language Model Activations), a probing-based method that leverages hidden layer activations of base LLMs to predict whether a statement is true or false. Crucially, the LLM itself is not fine-tuned; instead, a lightweight classifier is trained on internal representations extracted while the model reads or generates statements.

Through carefully designed experiments using topic-held-out evaluation, the authors demonstrate that SAPLMA significantly outperforms few-shot prompting and sentence-probability-based baselines. Accuracy levels between 60% and 80% are achieved for OPT-6.7B, and between 70% and 90% for LLaMA2-7B, compared to a maximum of approximately 56% for prompting-based approaches. These results indicate that truthfulness is encoded in the LLM’s internal states in a manner that generalizes across topics.

A key finding of the paper is that the internal representation of truthfulness is fundamentally different from the probability assigned to a sentence by the LLM. Sentence probabilities are strongly influenced by syntactic structure, token frequency, sentence length, and fluency, whereas SAPLMA’s predictions are far more robust to these factors. Qualitative examples further show that SAPLMA can correctly identify falsehoods even when they are linguistically common, as well as recognize true statements despite typographical errors or rare wording.

Overall, the paper provides strong empirical evidence that hallucination is often not due to a lack of internal knowledge, but rather a consequence of generation dynamics. *By exposing latent belief signals within LLMs, SAPLMA opens a path toward self-reviewing language models that can flag unreliable statements before they reach the user, thereby improving trust, transparency, and safety in LLM-based systems.*

10 Limitations and Future Directions

Limitations include:

- Binary truth labels
- English-only data
- Dependence on access to internal states

Future work includes:

- Scaling to larger LLMs
- Human-in-the-loop evaluations
- Studying activation dynamics during generation
- Multilingual extensions